

COLVILLE LAKE, NT, Canada

WMO: 710550

Lat: 67.0214N

Lon: 126.1231W

Elev: 259

StdP: 98.25

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.7	(c) -39.5	(d) -46.0	(e) 0.0	(f) -41.4	(g) -43.8	(h) 0.1	(i) -39.3	(j) 9.7	(k) -19.7	(l) 8.8	(m) -18.9	(n) 0.9	(o) 200	(p) 0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.3	(c) 24.3	(d) 15.5	(e) 22.2	(f) 14.8	(g) 20.2	(h) 13.9	(i) 16.9	(j) 22.3	(k) 15.7	(l) 20.6	(m) 14.6	(n) 19.1	(o) 4.2	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	10.8	18.8	13.5	10.0	18.0	12.3	9.2	16.9	48.6	22.3	45.0	20.8	41.9	19.2	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 8.4	(c) 7.3	(e) -44.2	(f) 28.2	(g) 3.0	(h) 1.9	(i) -46.3	(j) 29.5	(k) -48.1	(l) 30.6	(m) -49.8	(n) 31.6	(o) -52.0	(p) 33.0		
(a) 9.5	(b) 8.4	(c) 7.3	(e) -43.8	(f) 18.2	(g) 2.5	(h) 1.7	(i) -45.6	(j) 19.4	(k) -47.0	(l) 20.5	(m) -48.4	(n) 21.4	(o) -50.2	(p) 22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.9	-26.3	-24.6	-21.0	-10.1	2.0	10.9	14.4	11.1	4.6	-5.1	-16.9	-23.3		
	DBStd	15.90	8.00	7.46	7.53	7.08	5.75	4.56	3.99	4.22	4.37	4.78	6.65	7.44		
	HDD10.0	6478	1127	969	961	602	254	44	9	38	169	467	806	1033		
	HDD18.3	9235	1385	1202	1219	852	507	226	131	227	413	725	1056	1292		
	CDD10.0	299	0	0	0	0	6	71	146	71	6	0	0	0		
	CDD18.3	14	0	0	0	0	0	2	10	2	0	0	0	0		
	CDH23.3	108	0	0	0	0	1	24	71	13	0	0	0	0		
	CDH26.7	16	0	0	0	0	0	3	12	1	0	0	0	0		
(14) Wind	WSAvg	3.2	2.4	2.5	2.6	3.0	3.4	3.7	3.9	4.1	4.0	3.5	2.8	2.3		
(15) Precipitation	PrecAvg	328	19	21	14	12	20	38	50	51	37	25	22	18		
	PrecMax	504	39	120	34	39	62	70	88	82	76	41	49	35		
	PrecMin	240	3	5	2	1	2	11	21	22	22	13	9	5		
	PrecStd	55	9	20	8	9	11	16	19	18	12	8	9	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.2	-4.4	1.0	9.4	20.8	26.3	28.3	25.0	17.7	8.2	-0.1	-5.2		
		MCWB	-3.2	-5.3	-0.9	4.7	11.5	15.3	17.6	16.4	11.4	5.3	-0.4	-5.5		
	2%	DB	-7.5	-8.4	-3.8	5.5	17.0	23.1	25.2	22.1	14.7	3.8	-3.8	-8.3		
		MCWB	-7.9	-8.9	-5.0	2.3	9.6	14.0	16.2	15.7	10.0	2.6	-4.2	-8.6		
	5%	DB	-12.2	-11.4	-7.2	3.0	13.6	20.7	23.0	19.7	12.7	1.7	-5.7	-10.4		
		MCWB	-12.6	-11.8	-8.0	0.6	7.6	12.8	15.3	14.5	9.1	0.8	-6.1	-10.7		
	10%	DB	-15.7	-14.1	-10.2	0.6	10.6	18.5	21.1	17.5	10.7	0.4	-7.7	-12.5		
		MCWB	-16.0	-14.5	-10.9	-1.0	6.0	12.0	14.7	13.3	8.2	-0.3	-8.0	-12.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	-5.3	-0.8	5.0	11.7	16.2	19.1	17.7	12.6	5.5	-0.3	-5.5		
		MCDB	-2.3	-4.5	0.8	9.3	20.1	23.7	25.2	23.1	15.5	7.4	-0.1	-5.1		
	2%	WB	-7.9	-8.9	-5.0	2.5	9.7	14.8	17.5	16.3	10.9	2.6	-4.1	-8.6		
		MCDB	-7.5	-8.5	-3.9	5.3	16.3	21.1	23.2	20.7	13.8	3.7	-3.8	-8.4		
	5%	WB	-12.5	-11.8	-8.0	0.8	8.0	13.7	16.5	14.9	9.5	0.9	-6.1	-10.7		
		MCDB	-12.2	-11.4	-7.3	2.6	13.1	19.5	21.6	19.0	12.1	1.5	-5.7	-10.5		
	10%	WB	-16.0	-14.5	-11.0	-1.1	6.1	12.4	15.4	13.7	8.3	-0.2	-8.0	-12.8		
		MCDB	-15.7	-14.1	-10.3	0.7	10.5	17.6	19.9	17.2	10.7	0.4	-7.7	-12.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.3	10.1	11.4	9.6	9.6	8.3	7.2	5.8	4.7	6.8	7.1		
		MCDBR	10.5	9.9	12.6	12.2	13.8	13.1	11.3	10.4	9.3	5.1	7.0	8.5		
		MCWBR	10.3	9.6	11.2	9.3	8.1	6.5	5.3	5.5	5.5	4.2	6.9	8.3		
	5% WB	MCDBR	10.6	9.8	12.5	11.2	13.6	11.9	9.8	9.7	8.5	4.8	6.9	8.4		
		MCWBR	10.3	9.5	11.1	8.5	8.2	6.3	5.1	5.5	5.4	4.0	6.8	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.148	0.208	0.252	0.293	0.331	0.338	0.355	0.331	0.291	0.236	0.153	N/A		
	taud		1.932	2.218	2.231	2.200	2.340	2.451	2.426	2.505	2.563	2.284	1.935	N/A		
	Ebn at Noon		263	684	818	863	865	869	838	826	787	619	222	0		
	Edh at Noon		12	44	76	105	103	95	95	79	58	40	11	0		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.66	2.40	4.70	5.84	6.14	5.19	3.63	1.98	0.62	0.13	0.00		
	RadStd		0.01	0.05	0.10	0.21	0.45	0.47	0.39	0.33	0.19	0.07	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.89	+2.07	+2.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.70	+1.39	+1.33	N/A	N/A	N/A	-220	-203	N/A	N/A

Nomenclature: See separate page